Autonomous Physics-Based Modeling of Retention Loss in 3-D NAND Flash Memories

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***Abstract*— A novel model is established with physical derivation in retention operation of 3-D NAND flash memory. To efficiently incorporate the retention loss properties of the immediate and continuous occurance, an autonomous model has been suggested. By utilizing the model, retention loss for both short-term and long-term due to detrapping effect of trapped electron can be simulated in very short timeframe. Moreover, autonomous model makes it possible for model to predict the time transient of ΔV_{th} by only using an initial condition.**

***Index Terms*— Bandgap-Engineered 3D NAND Flash, Soft Program, Pass Voltage, ΔV_T Distribution**

I. Introduction

THERE is a rapid increase in the demand for data storage driven by modern computing technologies such as artificial intellegence (AI) and high-performance computing (HPC), 3-D NAND flash memories' reliability issue for retention loss must be concerned. As data storage capacity per cell increases, the reliability of each bit's ΔV_{th} distribution must be maintained over a long period, ranging from a few days to several years.

It has been reported that detrapping effect occur under cryogenic condition. This aligns with the previous study of detrapping modeled independent of temperature.

In this study, an autonomous modeling of the retention loss by tunneling emission of 3-D NAND flash memory has been constructed. Only initial condition matters for this simulation with highly fast simulation time.

II. Modeling

The modeling of autonomous simulation for retention loss is held in this section. First, a cylindrical poisson solution for O/N/O layer will be done with quantized multiple layer in CTN layer. After the detailed poisson solution for multiple quantized layers is solved, ΔV_{th} will be solved with optimized calculation. Last, based on a previous study of emission rate by detrapping effect, an autonomous equation is suggested.

A. Detailed Poisson Solution

An electrical potential distribution of a cylindrical Poisson equation for split layers in ONO structure can be solved as following equation.

$$\begin{cases} V_{\text{Si}}(r_{\text{Si}}) = V_{\text{ch}} \\ V_{\text{TOX}}(r) = V_{\text{ch}} + \frac{K_{\text{TOX}}}{\epsilon_{\text{TOX}}} \ln \frac{r}{r_{\text{Si}}} \\ V_{\text{CTN},i}(r) = V_{\text{CTN},i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{\text{CTN}}} \ln \frac{r}{r_{i-1}} \\ V_{\text{BOX}}(r) = V_{\text{CTN},N}(r_N) + \frac{K_{\text{BOX}}}{\epsilon_{\text{BOX}}} \ln \frac{r}{r_N} \\ V_{\text{BOX}}(r_{\text{BOX}}) = V_{\text{G}} \end{cases} \tag{1}$$

Solving $\mathbf{x}$ vector can calculate the total $N+2$ potential parameters.

$$\mathbf{x} = \begin{bmatrix} K_{TOX} & K_1 & \cdots & K_N & K_{BOX} \end{bmatrix}^\top \tag{2}$$

$$A\mathbf{x} = \mathbf{b} \tag{3}$$

A and b can be defined with total $N+1$ transverse electric field equality boundary conditions ($\epsilon_1 \mathcal{E}_1 = \epsilon_2 \mathcal{E}_2$) and one extra boundary condition with reference voltage.

$$A = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{\text{TOX}}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\text{BOX}}} \end{bmatrix} \tag{4}$$

$$\mathbf{b} = \begin{bmatrix} \frac{\Delta Q_0}{2\pi} & \frac{\Delta Q_1}{2\pi} & \cdots & \frac{\Delta Q_N}{2\pi} & \Delta V \end{bmatrix}^\top \tag{5}$$

$$\Delta Q_i = \pi r_i^2 \left(n_{t,i} - n_{t,i-1} \right) \tag{6}$$

$$\Delta V = (V_{\text{G}} - V_{\text{ch}}) - \sum_{i=1}^{N} \frac{q n_{t,i}}{4\epsilon_{\text{CTN}}} \left(r_i^2 - r_{i-1}^2 \right) \tag{7}$$

C_{TOX}, C_i, and C_{BOX} in (4) indicates capacitance of each cylindrical layers as shown in Fig. 1(b)-(c). ΔQ_i in (5) stands for a compenet of radius r_i cylindrical charge which can be superpositioned as $Q = \sum \Delta Q_i$ as demonstrated on Fig. 1(d). (5) shows that $\mathbf{b}(\Delta Q_i, V_{\text{G}} - V_{\text{ch}})$ is varied by trapped charge distribution and given voltage conditions.

B. Q - ΔV_{th} Relation

For ΔV_{th} calculation, gate voltage difference for equal channel field condition should be solved.

$$F_{\text{TOX}}(r_{\text{Si}}) = \frac{K_{\text{TOX}}}{\epsilon_{\text{TOX}}} \frac{1}{r_{\text{Si}}} \tag{8}$$

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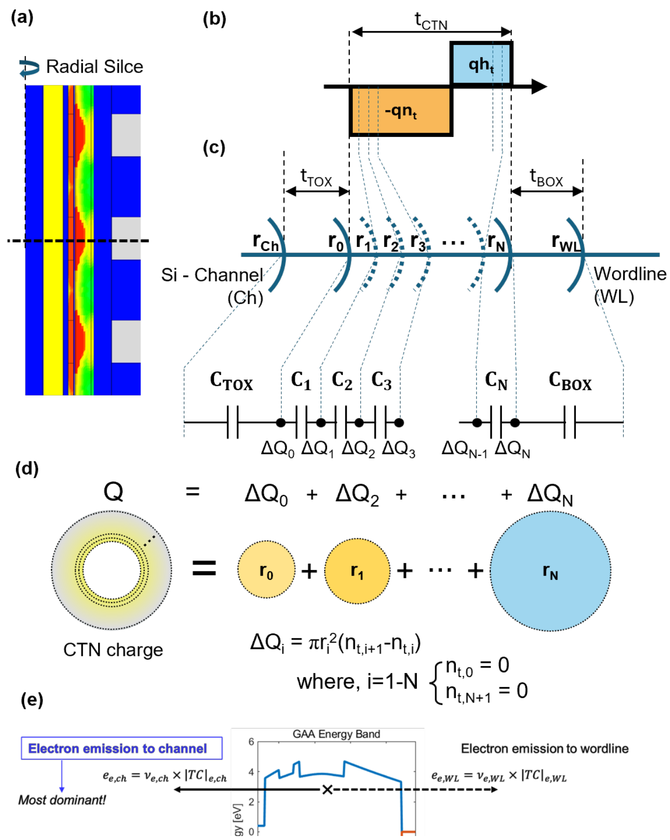

Fig. 1. (a) shows a radial scale and capacitance visulization, (b) shows the representation of $\Delta \boldsymbol{Q_i}$ within the boundary condition solution, (c) shows the programming sequence of trapped electrons and holes, and (d) shows the autonomous simulation schematics.

to achieve equal channel field, same K_{TOX} value under given conditions mush be established. By Cramer's rule, K_{TOX} parameter can be solved by determinant calculation.

$$K_{\text{TOX}} = \frac{\det\left[A_{\mathbf{b}(\Delta Q_i, \Delta V_{\text{th}})}\right]}{\det A} \tag{9}$$

where notation $A_{\mathbf{b}(\Delta Q_i, \Delta V_{\text{th}})}$ stands for 1st column of matrix A replaced with vector $\mathbf{b}(\Delta Q_i, \Delta V_{\text{th}})$.

Therefore, establishing following equation and solving for ΔV_{th} is needed to calculate the value.

$$\det\left[A_{\mathbf{b}(0,\Delta V_{\text{th}})}\right] = \det\left[A_{\mathbf{b}(\Delta Q_i,0)}\right] \tag{10}$$

by property of multilinearity, $\det\left[A_{\mathbf{b}(0,\Delta V_{\text{th}})}\right]$ in equation (10) can be calculated as following.

$$\det\begin{bmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \Delta V_{\text{th}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\text{BOX}}} \end{bmatrix} = -\Delta V_{\text{th}} \tag{11}$$

$$\therefore \Delta V_{\text{th}} = -\det\left[A_{\mathbf{b}(\Delta Q_i,0)}\right] \tag{12}$$

The threshold voltage difference can be calculated with single determinant computation with initially programmed trapped charge distribution is given as Fig. II-A(d).

TABLE I
PARAMETER VALUES FOR THE TCAD SIMULATION

Symbol	Physical meaning	Value
r_f	Oxide filler radius	20 nm [?]
t_{Si}	Si-channel thickness	8 nm [?]
t_{TOX}	Tunneling oxide thickness	5 nm [?]
t_{CTN}	Charge trap nitride thickness	6 nm [?]
t_{BOX}	Blocking oxide thickness	8 nm [?]
ε_{ox}	Oxide permittivity	3.9 ε_0 [?]
ε_{CTN}	Charge trap nitride permittivity	7.5 ε_0 [?]
$\Delta\Phi_{\text{Si-ox}}$	Si-ox layer energy barrier height	3.1 eV [?]
$\Delta\Phi_{\text{CTN-ox}}$	CTN-ox layer energy barrier height	1 eV [?]
$\Delta\Phi_{\text{WL-ox}}$	WL-ox layer energy barrier height	3.65 eV [?]
m^*_{ox}	The effective mass of oxide	0.5 m_0 [?]
m^*_{n}	The effective mass of nitride	0.4 m_0 [?]
σ_{n}	electron capture cross section	1×10^{-16} cm^2 [?]
V_{pass}	Pass voltage for untarget cell	7.0 V [?]
$V_{\text{pgm\_0}}$	Program voltage for ΔV_{th} = 0 V	18.6 V
$V_{\text{pgm\_1}}$	Program voltage for ΔV_{th} = 1 V	19.1 V
$V_{\text{pgm\_2}}$	Program voltage for ΔV_{th} = 2 V	19.6 V

C. Autonomous Detrapping Modeling

With given initial condition, an autonomous equation for each cell divided be radial and trap depth as Fig. II-A(e) is established.

$$\frac{d}{dt} n_t(r, E_t, t) = -(e_{\text{ch}} + e_{\text{WL}}) n_t(r, E_t, t) \tag{13}$$

Then, detrapping relation can be express as following general ODE solution.

$$n_t(r, E_t, t) = \sum_r \sum_{E_t} n_{t0}(r, E_t) \exp(-(e_{\text{ch}} + e_{\text{WL}})t) \tag{14}$$

for hole concentration (h_t) vice versa.

The emission rate for trapped electron and hole can be computed as following equations based on a previous study [1].

$$e = \nu \times |TC| \tag{15}$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\text{trap}} \cdot \frac{m * \sqrt{2m*}}{\pi^2 \hbar^3} \sqrt{E - E_{\text{trap}}} \tag{16}$$

$$|TC| = \prod \exp \frac{4\sqrt{2m^*}}{3\hbar q} \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F} \tag{17}$$

for both channel and wordline directions as demonstrated in Fig. 2(a).

This autonomous detrapping modeling methodology enables a simulation to directly comput the result at a specific time without performing any transient time-interval calculations. Only the initial condition and the target time are required, eliminating the need for time-stepped simulations.

III. Simulation and Result

The validation of the model is done with Synopsys TCAD simlation which is calibrated with previous studies of retention modeling [?], [1]. In this study, to validate the model properly focused on the scope of detrapping effect from CTN layer, bandgap-engineered process was neglected and the operation conditions such as program voltage were calibrated again for this research.

After the erase operation, 1 μs of program voltage was given to all the wordlines of the NAND as table I. Note that ΔV_{th}

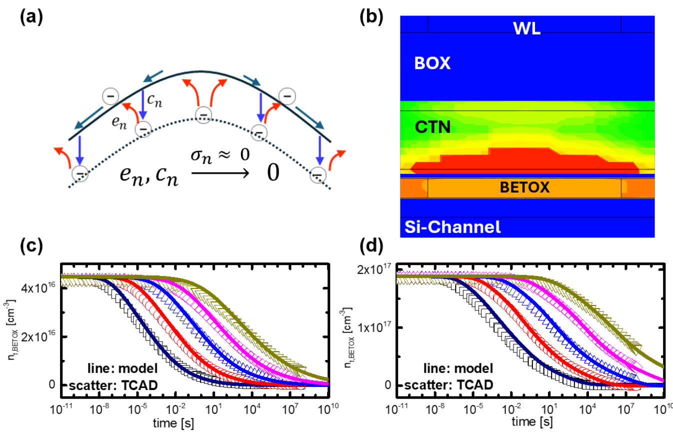

Fig. 2. (a) shows a radial scale and capacitance visulization, (b) shows the representation of ΔQ_i within the boundary condition solution, (c) shows the programming sequence of trapped electrons and holes, and (d) shows the autonomous simulation schematics.

meaning the threshold voltage shift from the virgin cell, which starts from negative value in erased state.

According to the previous study [2], one can achieve this separation of detrapping mechanism by giving cryogenic condition. In order to reproduce the cryogenic condition in retention operation, authors have reached this condition by manipulating the capture cross section to extremely low value ($\sigma_\mathrm{n} \rightarrow 0$) only at the retention operation. This manipulation suppresses the thermal transition of electrons between trapped and conduction state within CTN layer as described at Fig. III(a). This parameter tuning allowed the simulation to emulate the cryogenic effect within the TCAD framework for the retention operation, while maintaining physical consistency in the other operations.

The band diagram model and TCAD simulation setup of programmed voltage shows good match as Fig. II-A(e).

The setup of the simulation was first done with an erase operation, filling up the CTN layer full of holes with sufficient erase operation time [**?**], [**?**]. First, validation over time span to testing time of 10 years is done as Fig. III(a). This result shows that the modeling done with the model stated within the previous section very well matches with the referenced TCAD simulation with less than 1

A validation for autonomous modeling is done on Fig. III(c). Retention loss over time period makes the change of the band diagram. However, Fig. III(c) shows that even with several refreshments over time period, there are very small error compared to autonomous modeling with less than 1% of WMAPE. Therefore, autonomous simulation can save computational resource and improve calculation time.

IV. Conclusion

An autonomous modeling of retention loss due to detrapping of trapped electron has been done in the study. Compared to TCAD simulation with physical parameters calibrated to various previous studies, the model showed its accuracy by achieving less WMAPE less than 1%. Also, novelty of autonomous modeling has been also shown that one can reach a goal of computational optimization within extremely short timeframe.

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